

Question Number : 109 Question Id : 640653455676 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

How many unique *du-paths* are there for the variable: x , y and z ?

Options :

6406531515190. ✓ 7

6406531515191. ✗ 8

6406531515192. ✗ 9

6406531515193. ✗ 10

Industry 4.0

Section Id :	64065329449
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	45
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065365951
Question Shuffling Allowed :	No

Is Section Default? : null

Question Number : 110 Question Id : 640653455679 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL: INDUSTRY 4.0"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECT TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT ,PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531515202.  Yes

6406531515203.  No

Sub-Section Number : 2

Sub-Section Id : 64065365952

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 111 Question Id : 640653455680 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Match the layout to the type of industry it can (most predominantly) be used in

Layout	Product
1. Fixed position layout	A. Packaged Bottled Drinks
2. Product layout	B. Cooking in a kitchen
3. Process layout	C. Construction

Options :

6406531515204. ✓ 1-C, 2-A, 3-B

6406531515205. ✗ 1-B, 2-C, 3-A

6406531515206. ✗ 1-A, 2-C, 3-B

6406531515207. ✗ 1-C, 2-B, 3-A

6406531515208. ✗ None of these

Question Number : 112 Question Id : 640653455689 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets on the neighbouring shelves. He picked a few milk-powder packets. However, these were not part of his list. He was searching for Good-day biscuits. He could identify only one packet on the shelf. However, there were a lot of Sun-feast biscuits lying on the next shelf. So, he picked ten Sun-feast packets. Now map the scenarios with the items below:

Product	Scenario
P1) Tea : Coffee	S1) Complementary
P2) Tea: milk-powder packet	S2) Substitution
P3) Good-day biscuits: Sun-feast biscuits	S3) Inventory dependant demand
	S4) Impulse buying

Options :

6406531515221. ✗ P1-S1, P2-S3, P3-S4

6406531515222. ✓ P1-S2, P2-S1, P3-S3

6406531515223. ✖ P1-S2, P2-S1, P3-S4

Question Number : 113 Question Id : 640653455690 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Location problem is not dependent on which of the following factors?

Options :

6406531515224. ✖ Distance

6406531515225. ✖ Number of facilities

6406531515226. ✖ Optimization criteria

6406531515227. ✔ None of these

Sub-Section Number : 3

Sub-Section Id : 64065365953

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 114 Question Id : 640653455693 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Which of them is not a part of the 7 principles of IoT:

Options :

6406531515234. ✖ Big Analog Data

6406531515235. ✔ Quality Control

6406531515236. ✖ Constant connection

6406531515237. ✖ Real time

6406531515238. ✖ Spectrum of insights

Sub-Section Number :

4

Sub-Section Id :

64065365954

Question Shuffling Allowed :

No

Is Section Default? :

null

Question Id : 640653455681 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (115 to 117)

Question Label : Comprehension

The ten activities specified below are to be performed to make a product. Then answer the given subquestions.

Activity	Activity Time (in min)	Preceding Activity
A1	9	
A2	5	
A3	18	
A4	10	
A5	6	
A6	8	A1, A3
A7	16	A1, A4
A8	19	A3, A5
A9	5	A2
A10	13	A8, A9

Table Indicating Activity Information

Sub questions

Question Number : 115 Question Id : 640653455682 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is/are an allowable grouping of activities to produce the product if the target is to produce a product every 20 minutes?

Options :

6406531515209. ✓ (A3) → (A1,A4) → (A2,A5,A6) → (A8) → (A9,A10) → (A7) → Output

6406531515210. ✖ (A3,A2) → (A1,A4) → (A5,A6) → (A8) → (A10,A9) → (A7) → Output

6406531515211. ✖ (A2,A5,A6) → (A1,A4) → (A3) → (A8) → (A10,A9) → (A7) → Output

6406531515212. ✖ (A3) → (A1,A4,A2) → (A8) → (A10,A9) → (A7) → Output

Question Number : 116 Question Id : 640653455683 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the balance delay for the line (or lines) identified in the previous question? *(Enter your answer in % rounded to two decimal places, without the percentage symbol. Eg. If you compute the answer as 0.10005, then enter 10.01 as your answer)*

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

9.10 to 9.20

Question Number : 117 Question Id : 640653455684 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

If a day has 2 shifts and each shift is for 5 hours, then what is the maximum output per day that can be achieved from the layout(s) chosen as per the previous question number 6?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Id : 640653455685 **Question Type :** COMPREHENSION **Sub Question Shuffling Allowed :** No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix **Calculator :** None **Response Time :** N.A **Think Time :** N.A **Minimum Instruction Time :** 0
Question Numbers : (118 to 119)

Question Label : Comprehension

The Bellman equation is solved using backward recursion for a 3-period inventory model. The ordering cost is Rs. 2500 per order; the inventory holding cost is Rs. 250 per unit per period; the material cost is Rs. 15,000 per unit, and the item is sold at Rs. 20,000 per unit. The inventory can hold at most 30 units at any given period. Then answer the given subquestions.

Sub questions

Question Number : 118 **Question Id :** 640653455686 **Question Type :** SA **Calculator :** None
Response Time : N.A **Think Time :** N.A **Minimum Instruction Time :** 0
Correct Marks : 4

Question Label : Short Answer Question

There are 10 units at the start of the second period and the demand in the period is 20 units. The possible future profit is Rs. 7,47,500 irrespective of how many units are in the inventory at the end of the second period. Then what is the maximum profit that can be generated up to period 2 (when the possible actions are restricted to ordering either 0, 10 or 20 units in period 2)?

(Hint: You are to choose the best profit using backward recursion) (Note: Enter the numerical answer in Rupees, without the "Rs." notation. Eg. If you get a profit of Rs. two thousand (Rs. 2000), enter the answer as 2000)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1195000

Question Number : 119 **Question Id :** 640653455687 **Question Type :** SA **Calculator :** None

Response Time : N.A **Think Time :** N.A **Minimum Instruction Time :** 0

Correct Marks : 2

Question Label : Short Answer Question

If demand in the second period is not deterministic and follows a distribution as given in the table below, then what is the probability, that period 2 will end with 0 units in inventory if period 2 starts with 10 units in inventory and an order is placed for 20 units?

(Hint: You need to find the transition probability that period-3 will begin with 0 units of inventory if period 2 starts with 10 units in inventory and an order of 20 units is placed in period 2)

Demand	Probability of Demand
10	0.25
20	0.25
30	0.25
40	0.25

(Note: Enter your answer in decimal values only.)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Question Id : 640653455694 **Question Type :** COMPREHENSION **Sub Question Shuffling**

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (120 to 121)

Question Label : Comprehension

Let us assume a flight with capacity $c = 100$ (identical seats), and the price for economy passengers is \$200, and the price for business passengers is \$600. If the demand for business passengers is uniformly distributed between $[10, 40]$, then answer the given subquestions.

Sub questions

Question Number : 120 Question Id : 640653455695 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

What is the protection level? (Round your answer to the nearest higher integer. For example, if the answer is 10.5 round it to 11)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

29 to 31

Question Number : 121 Question Id : 640653455696 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

What is the booking limit? (Round your answer to the nearest higher integer. For example, if the answer is 10.5 round it to 11)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

69 to 71

Question Id : 640653455706 **Question Type :** COMPREHENSION **Sub Question Shuffling**
Allowed : No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix
Calculator : None **Response Time :** N.A **Think Time :** N.A **Minimum Instruction Time :** 0
Question Numbers : (122 to 125)

Question Label : Comprehension

A company needs to ship two products (A and B) to a foreign country. Two models of transport (Air and Sea) are available for shipping. The demand in the foreign market is stable at 100 units per week for each of the two products. Products A and B cost Rs. 20,000 and Rs. 10,000 respectively. If the company chooses to use air as the mode of transport, then it must ship 100 units in every shipment. On the other hand, if Sea is chosen as the mode of transport, then the shipment size must be 400 units. Air shipments take 1 week to reach the destination while Sea shipments take 4 weeks. Items shipped by Air incur a cost of Rs. 360 per unit and those shipped by Sea incur Rs. 90 per unit. If the company incurs 20% of the per unit cost for a product as the cost to hold the item in inventory for a year, answer the given subquestions. (Assume that a year has 52 weeks)

Sub questions

Question Number : 122 **Question Id :** 640653455707 **Question Type :** SA **Calculator :** None
Response Time : N.A **Think Time :** N.A **Minimum Instruction Time :** 0
Correct Marks : 1

Question Label : Short Answer Question

When shipping Product-B by Air, what is the total inventory in the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

150

Question Number : 123 Question Id : 640653455708 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

When shipping Product-B by Sea, what is the total inventory in the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

400

Question Number : 124 Question Id : 640653455709 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the total cost that is incurred for the movement of Product-A by Air (enter your answer in Lakh of rupees without the rupee symbol and punctuations. E.g: If you get an answer as "1 Lakh", then enter 100000)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2472000

Question Number : 125 Question Id : 640653455710 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the total cost that is incurred for the movement of Product-A by Sea (enter your answer in Lakh of rupees without the rupee symbol and punctuations. E.g: If you get an answer as "1 Lakh", then enter 100000)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2868000

Sub-Section Number : 5

Sub-Section Id : 64065365955

Question Shuffling Allowed : No

Is Section Default? : null

Question Id : 640653455697 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (126 to 127)

Question Label : Comprehension

During COVID, a company had asked its salesman to visit a set of locations (L1 to L5) to carry out the business. Now, post-COVID, business is picking up. Hence, the company is planning on hiring more salesmen. However, all salesmen must come to the office (OF) every day before leaving for the different locations, and return back to the office (OF) at the end of the day. Given the distance matrix for the locations set as below, answer the given subquestions.

Location	L1	L2	L3	L4	L5
L1					
L2	7				
L3	9	6			
L4	8	5	9		
L5	10	6	8	6	
OF	4	3	5	2	5

Distance Matrix in Kilometres

Sub questions

Question Number : 126 Question Id : 640653455698 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Then, during COVID, which of the following route(s) would have been best for the salesman?

Options :

6406531515241. ✓ L1 → L3 → L4 → L2 → L5 → L1

6406531515242. ✓ L1 → L4 → L3 → L2 → L5 → L1

6406531515243. ✓ L2 → L3 → L4 → L1 → L5 → L2

6406531515244. ✗ None of these

Question Number : 127 Question Id : 640653455699 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Post-COVID, if a salesman should not travel more than 10 Kilometers in a day, then how many salesmen will have to be hired by the company to visit all the locations every day?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number : 6

Sub-Section Id : 64065365956

Question Shuffling Allowed : No

Is Section Default? : null

Question Id : 640653455702 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (128 to 130)

Question Label : Comprehension

300 units of an LCD TV are sold every day in a store. Each LCD TV is produced at a cost of Rs. 500. A fixed cost of Rs. 1000 is incurred for a lot (irrespective of the number of items in the lot) to produce it. 20% of the cost to produce the LCD TV is incurred to store the LCD TV in the inventory for one year. If the LCD TV is sold for 300 days a year.

Based on the above data and answer the given subquestions.

Sub questions

Question Number : 128 Question Id : 640653455703 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the economic order quantity for the LCD TVs?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

1341

1342

Question Number : 129 Question Id : 640653455704 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the average inventory that must be carried for the LCD TVs?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

670

671

Question Number : 130 Question Id : 640653455705 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

How many orders will be placed for LCD TVs?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

67

68

Sub-Section Number : 7
Sub-Section Id : 64065365957
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 131 Question Id : 640653455691 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following statements is/are true? Choose all that are applicable

Options :

- 6406531515228. ✓ Euclidean distance measures distance as the straight line between 2 points.
- 6406531515229. ✓ Metropolitan distance measures distance with respect to the road availability
- 6406531515230. ✗ None of these

Question Number : 132 Question Id : 640653455692 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Objective of the p-median problem is to _____. Choose all that are applicable

Options :

- 6406531515231. ✓ Minimize the total weighted distance travelled from all demand centers to the opened facilities
- 6406531515232. ✗ Maximize the service demand by identifying the budgeted facilities
- 6406531515233. ✗ None of these

Sub-Section Number : 8

Sub-Section Id :	64065365958
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 133 Question Id : 640653455700 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is Digitization?

Options :

- 6406531515246. ✓ Conversion of physical 2D drawings to CAD drawings
- 6406531515247. ✗ Creating PLC logic in a microprocessor-based system to process data and identify insights
- 6406531515248. ✗ Providing real-time maintenance schedule based on real-time machine performance
- 6406531515249. ✗ None of these

Question Number : 134 Question Id : 640653455701 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

What is Digitalization? Choose all that are applicable

Options :

- 6406531515250. ✓ Use of digital technologies to change the processes
- 6406531515251. ✓ Process of moving towards a digital business
- 6406531515252. ✗ Pure Analog to Digital Conversion
- 6406531515253. ✗ Mere transformation – encoded in digital format